Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/20/2006
PI Name:	Bers, Donald M Ph.D.		
Project Title:	Integrative Cardiac Myocyte Model: Ion Channels, Ca and Contraction		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	dbers@lumc.edu	**Fax:**	FY 708-216-6308
PI Organization Type:	UNIVERSITY	**Phone:**	708-216-1018
Organization Name:	Loyola University, Chicago		
PI Address 1:	Department of Physiology		
PI Address 2:	2160 South First Avenue		
PI Web Page:	http://www.luhs.org/		
City:	Maywood	**State:**	IL
Zip Code:	60153	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	08/01/2001	**End Date:**	07/31/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	CA.002.14 IHF.002.04		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Our long-term objective is to build a detailed quantitative model of cardiac muscle, which will include electrophysiological properties, cellular Ca-regulation and contractile activation and relaxation. Aims of the project are: 1) To develop a more up-to-date electrophysiological model of cardiac myocyte dynamics; 2) to incorporate new Ca-transport data on SR Ca-uptake, release and Na/Ca exchange; 3) To extend the model to include cooperative Ca-dependent myofilaments activation, contraction and relaxation; and 4) Implement this model on highly accessible formats (one user-friendly and one computer-friendly for interfacing with other models). The rationale for our proposal stems from a lack of integration of information from the level of ion channels to Ca transients to myofilament force and shortening. We plan to develop a comprehensive quantitative model that incorporates up-to-date information on ion channels, modulation of Ca-cycling processes and myofilament activation. The approach involves a team of investigators with long-standing laboratory and modeling experience in each of these processes. Drs. Bers and Puglisi of Loyola University have extensive experience in modeling and quantitative experimental studies of regulation of		

Task Description:	Ca-cycling and membrane currents in the cardiac myocyte. Similarly, Drs. De Tombe and Solaro of the University of Illinois at Chicago have extensive experimental and modeling experience that focuses on the modulation of myocyte response to Ca. The approach involves the generation of computer models constrained by dynamic data generated in these and other laboratories. A preliminary working model LabHEART4 shows feasibility and utility of the proposed modeling format. This endeavor will provide new insights into normal physiological regulation of myocyte activity as well as providing a baseline from which altered function induced by alterations in hemodynamic loading can be readily considered. A fully integrated model of cardiac myocyte activity and regulation of myocyte activity developed here will have a broad application to acute changes in the environment as occurs in space travel and re-entry. This understanding will also have a broad significance to the understanding the effects of altered gene and protein expression associated with long-term changes in cardiac loading that occur during space travel and return to Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work has direct benefits of incorporating a tremendous amount of detailed experimental work into critically evaluated mathematical model (developed by leaders in the relevant experimental fields). This will allow others to both learn from and test our model. This will give investigators in many related fields a clearer understanding of how the heart works at the cellular level. The wide distribution and interest also means that we are likely to get valuable feedback from other scientists. Where the model is imperfect (inevitable), it will show us what is inadequate in our understanding and allow us to refine the mathematical description. This quantitative approach to understanding how the heart works is an important step toward realizing our overall goals to understand the workings of the heart in sufficient detail to make meaningful predictions in a very complex system. This has intrinsic benefits to society and public health, and how adjustments are made in a variety of non-physiological situations (pathophysiology or abnormal environments).
Task Progress:	No progress this reporting period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)